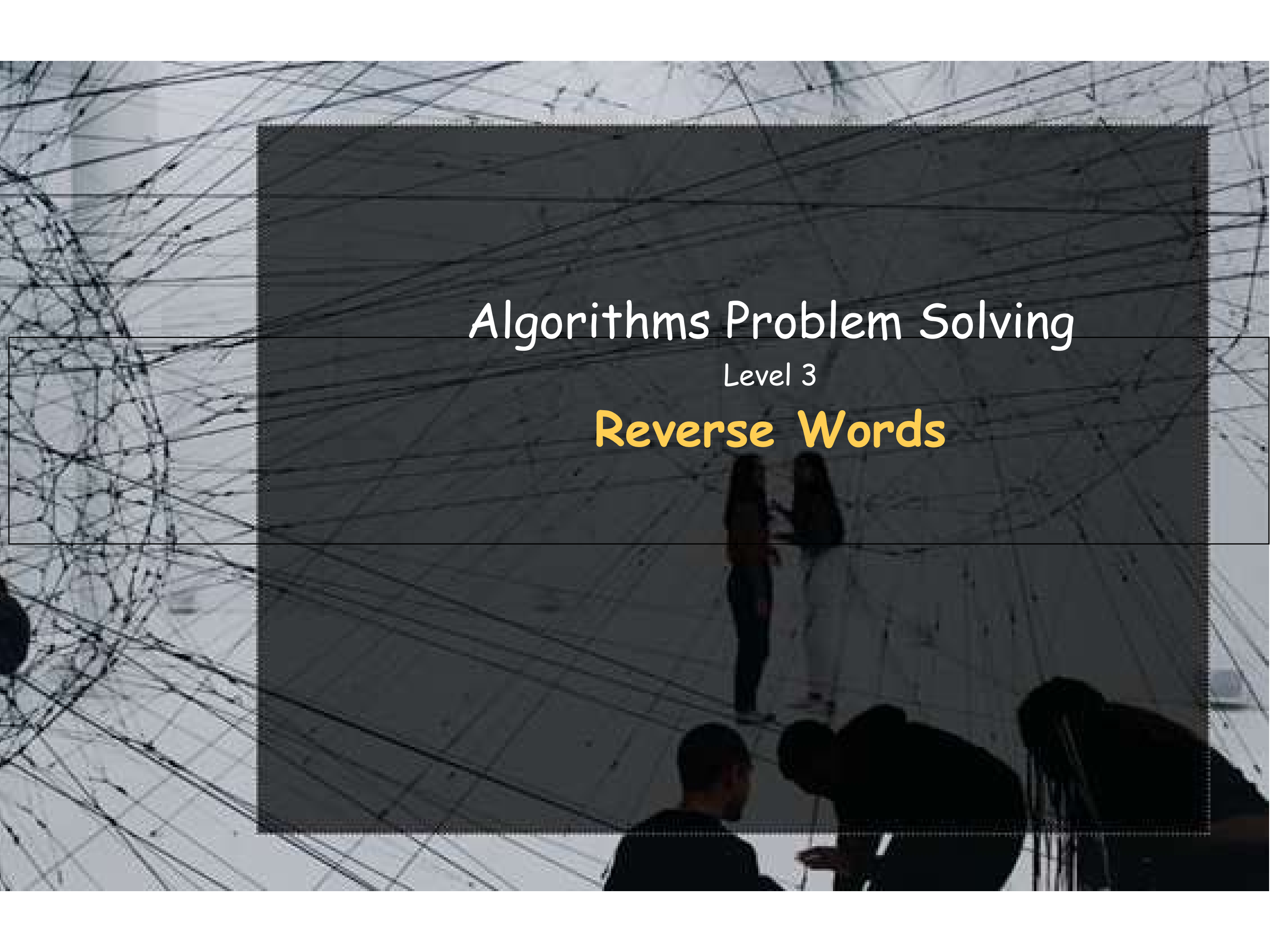


Algorithms Problem Solving

Level 3

Reverse Words



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Definition

Word Order Reversal

 Program reverses the order of words in a sentence while keeping the characters inside each word unchanged.

 For example, `"Hello World"` becomes `"World Hello"`.

Explanation

Tokenization

-  The sentence is first divided into individual words using ``SplitString()`.`

-  This converts the problem from character manipulation into vector element manipulation.

Reverse Traversal

-  The vector is processed from the last word toward the first.

-  The original spelling of each word remains unchanged.

Sentence Reconstruction

-  The reversed tokens are joined using spaces.

-  A new sentence is produced in reverse word order.

Procedures

1. Read the sentence.
2. Split it into a vector of words.
3. Start from the last vector element.
4. Traverse toward the first element.
5. Append each word to the result.
6. Insert spaces between words.
7. Display the reversed sentence.

💡 Example

Input:

Sara Alterkawi

➡ Result

Alterkawi Sara

Note

- ◆ Reversing words is different from reversing all characters.
- ◆ `std::reverse()` can reverse vector elements directly.
- ◆ `JoinString()` can be reused to reconstruct the final sentence.

⊘ Warning

- ◆ Do not confuse vector reversal with string character reversal.
- ◆ Unsigned `size()` values require care when looping backward to avoid underflow.
- ◆ Multiple spaces may be normalized depending on how `SplitString()` handles empty tokens.



Comparison

`ReverseWordsInString()`

- ✓ Reverses word order.
- ✓ Keeps characters inside each word unchanged.
- ✓ Uses tokenization and reassembly.

`ReverseString()`

- ✓ Reverses every character.
- ✓ Also reverses the characters inside each word.
- ✓ Produces a completely different result.

✦ Summary

- ✓ Splits a sentence into individual words.
- ✓ Traverses the words in reverse order.
- ✓ Preserves each word's internal spelling.
- ✓ Reconstructs a new sentence with reversed word positions.

Thank you 😊